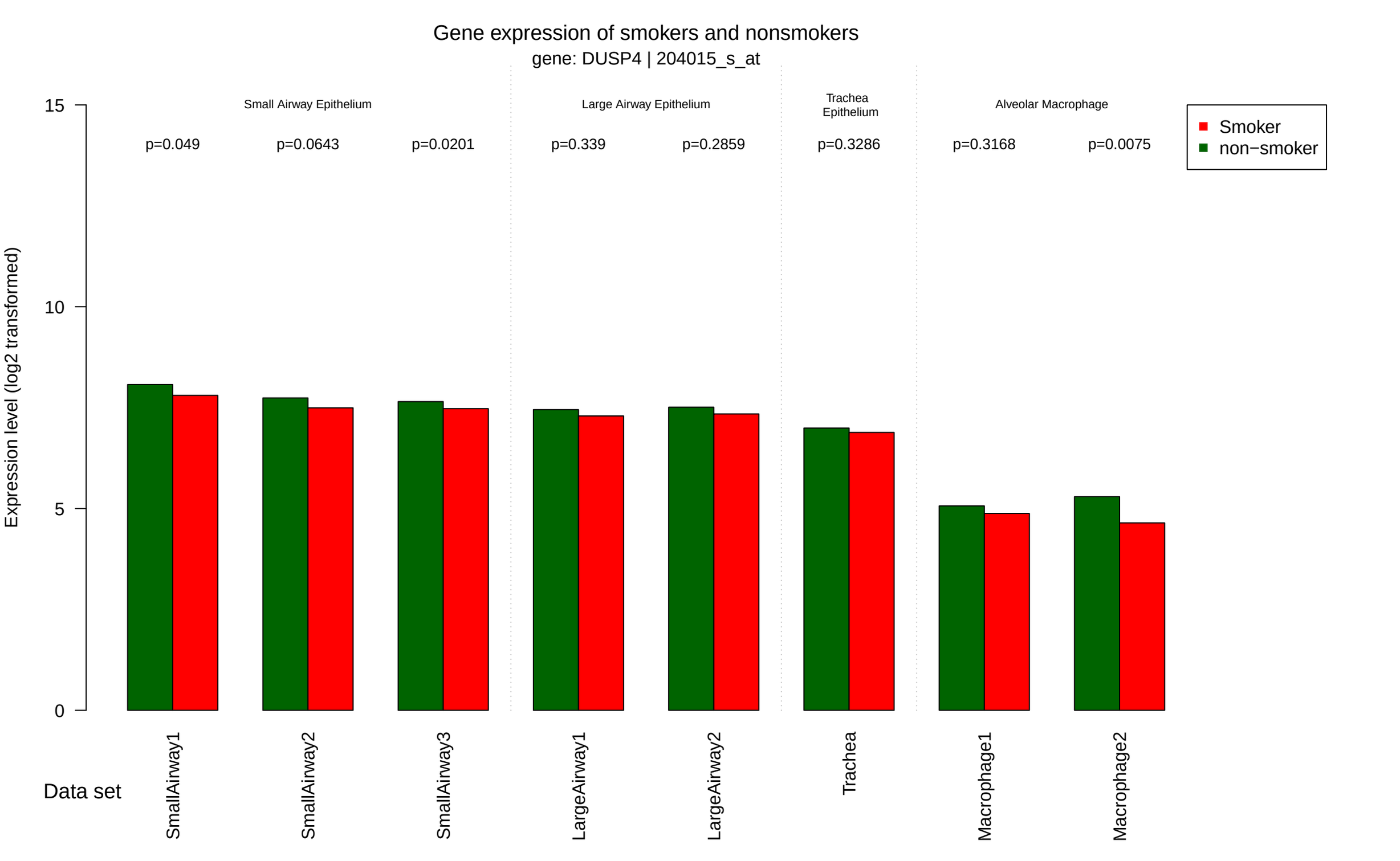
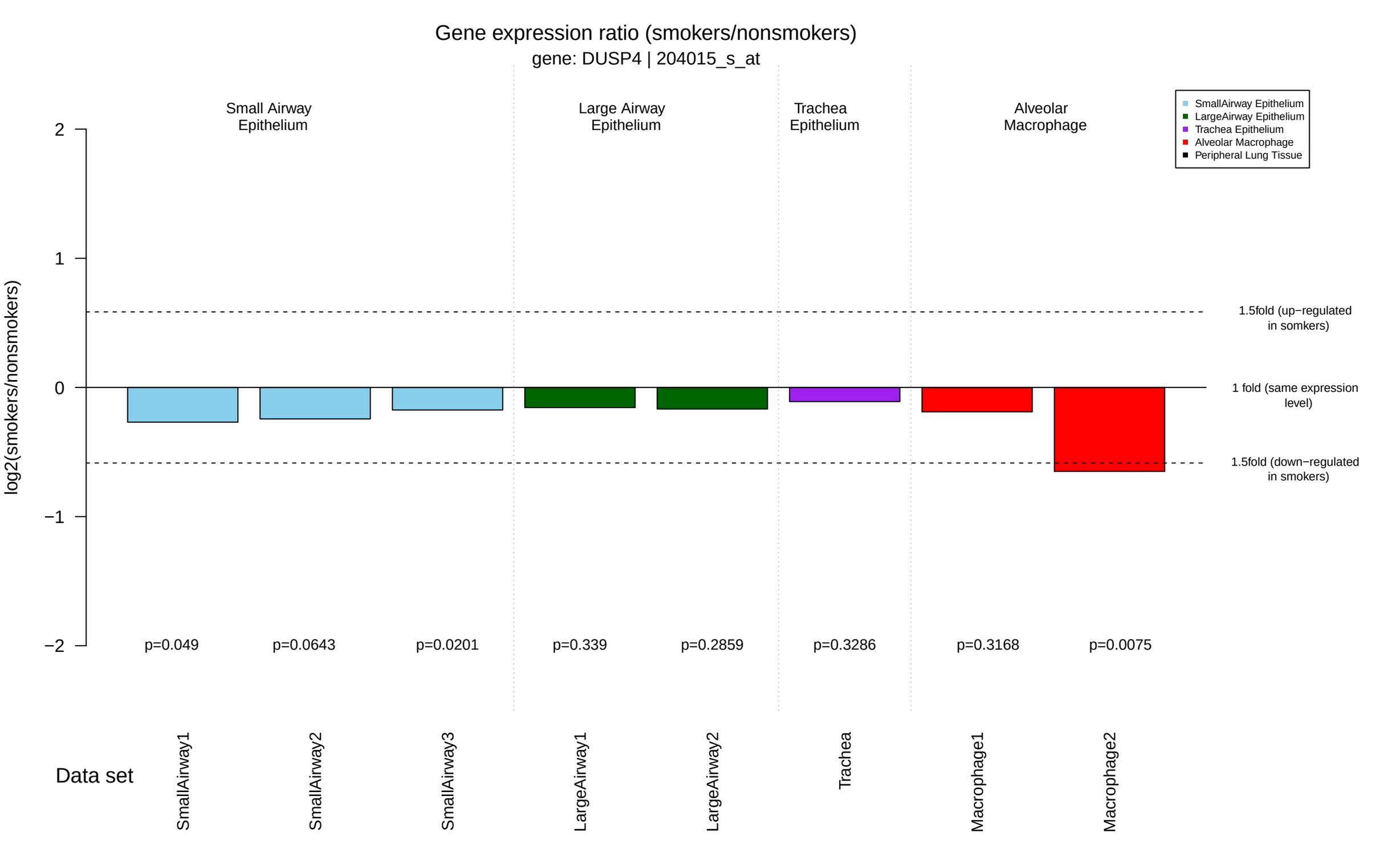
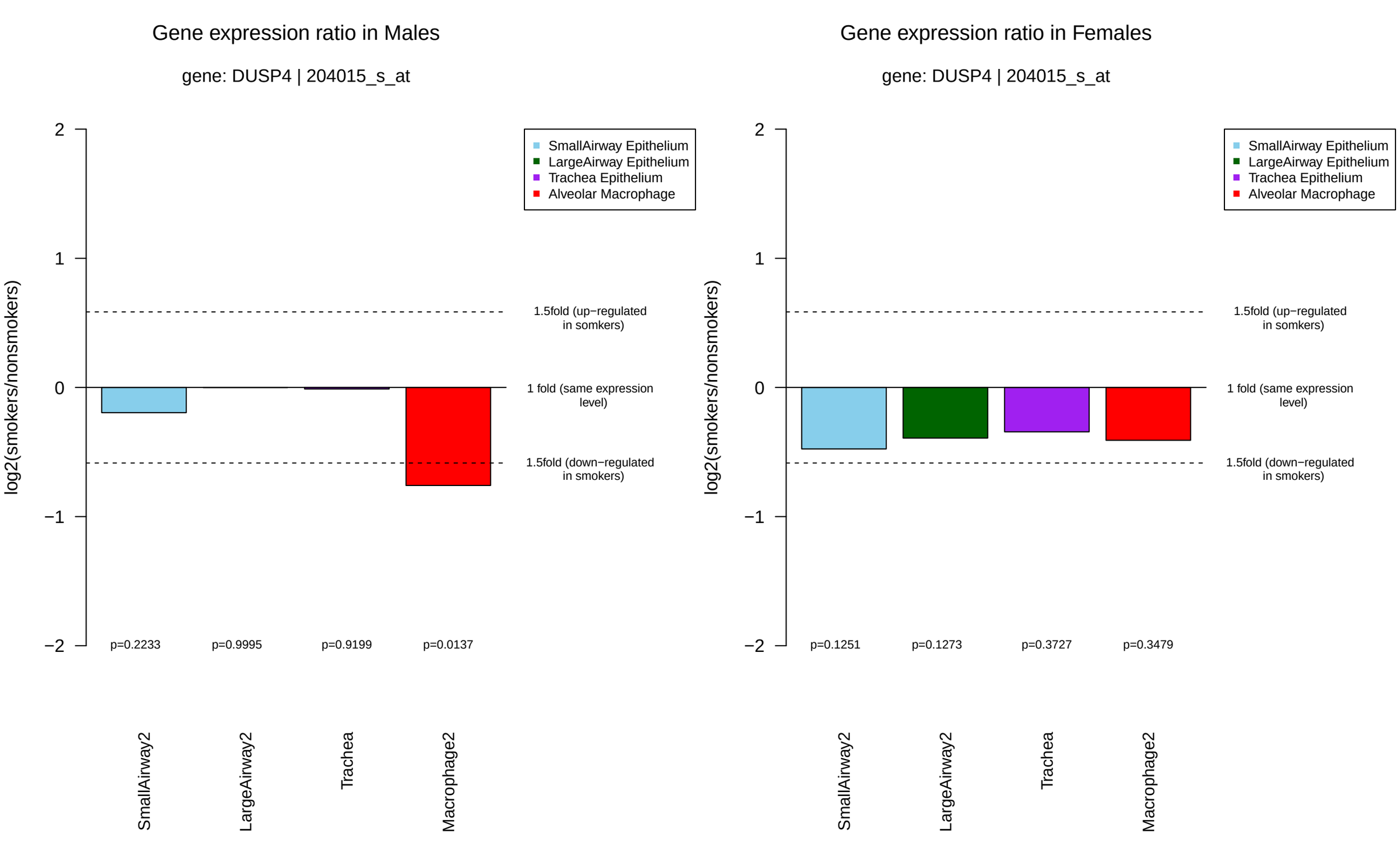
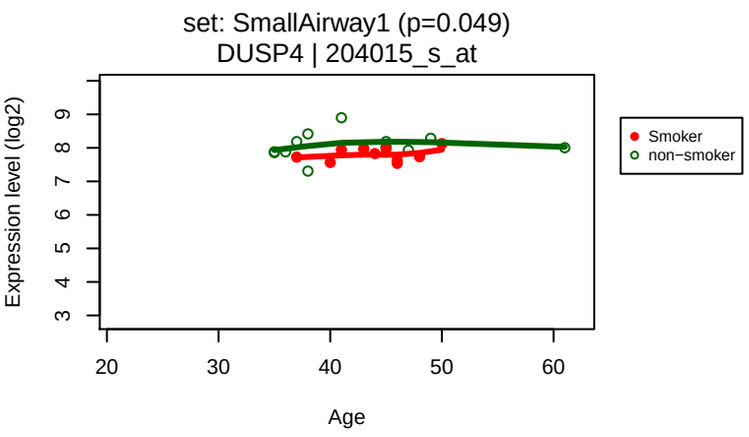
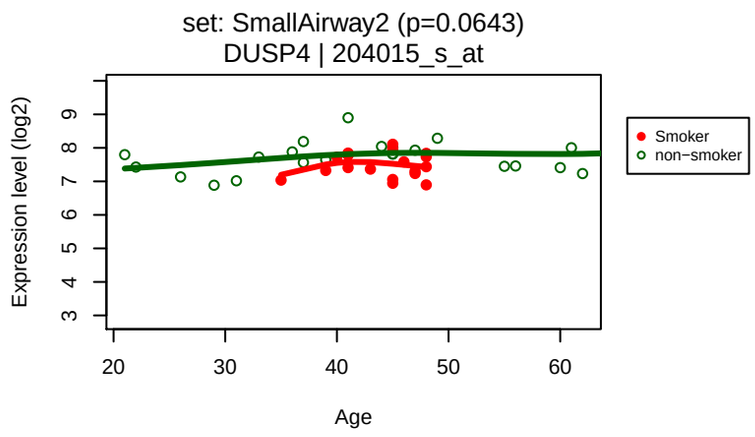
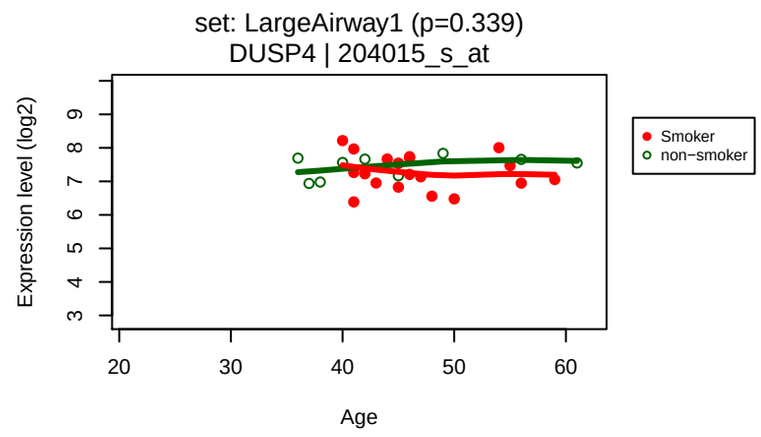

Gene expression of smokers and nonsmokers
gene: DUSP4 | 204015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.049
p=0.0643
p=0.0201
p=0.339
p=0.2859
p=0.3286
p=0.3168
p=0.0075
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

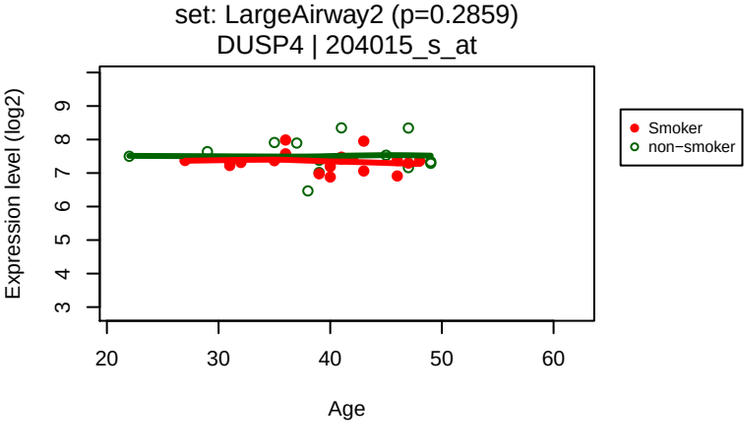

Gene expression ratio (smokers/nonsmokers)
gene: DUSP4 | 204015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.049
p=0.0643
p=0.0201
p=0.339
p=0.2859
p=0.3286
p=0.3168
p=0.0075
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

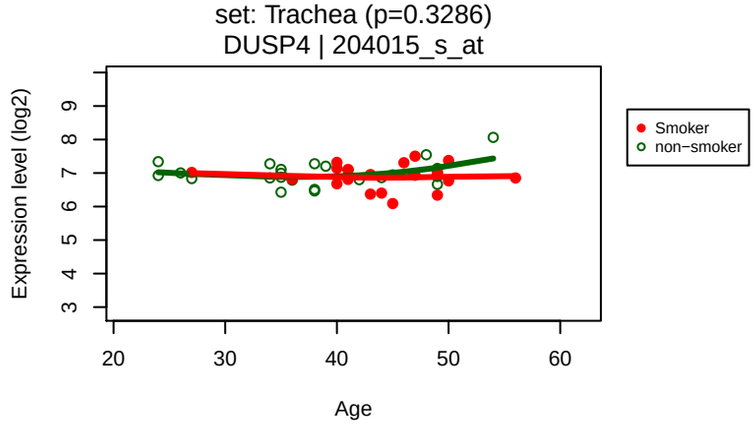

Gene expression ratio in Males
gene: DUSP4 | 204015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2233
p=0.9995
p=0.9199
p=0.0137
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DUSP4 | 204015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1251
p=0.1273
p=0.3727
p=0.3479
SmallAirway2
LargeAirway2
Trachea
Macrophage2

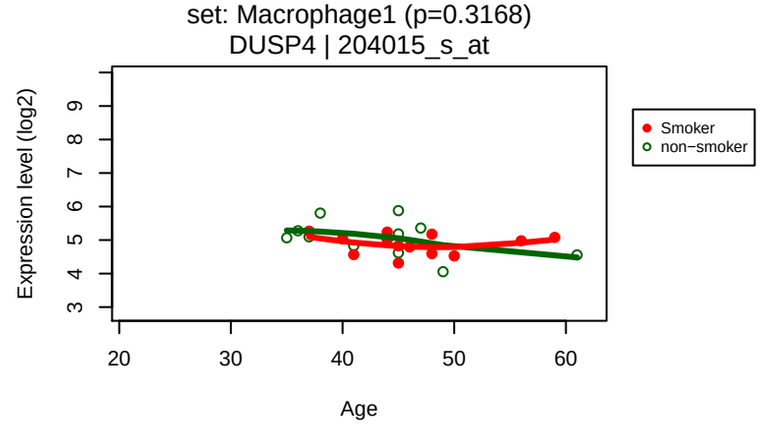

set: SmallAirway1 (p=0.049)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker

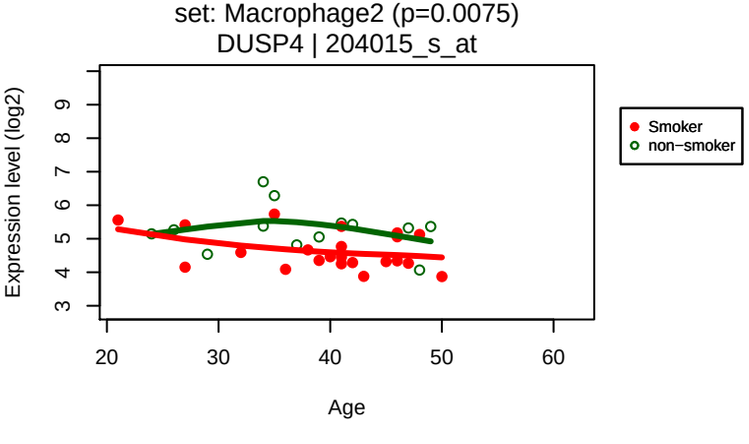

set: SmallAirway2 (p=0.0643)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker

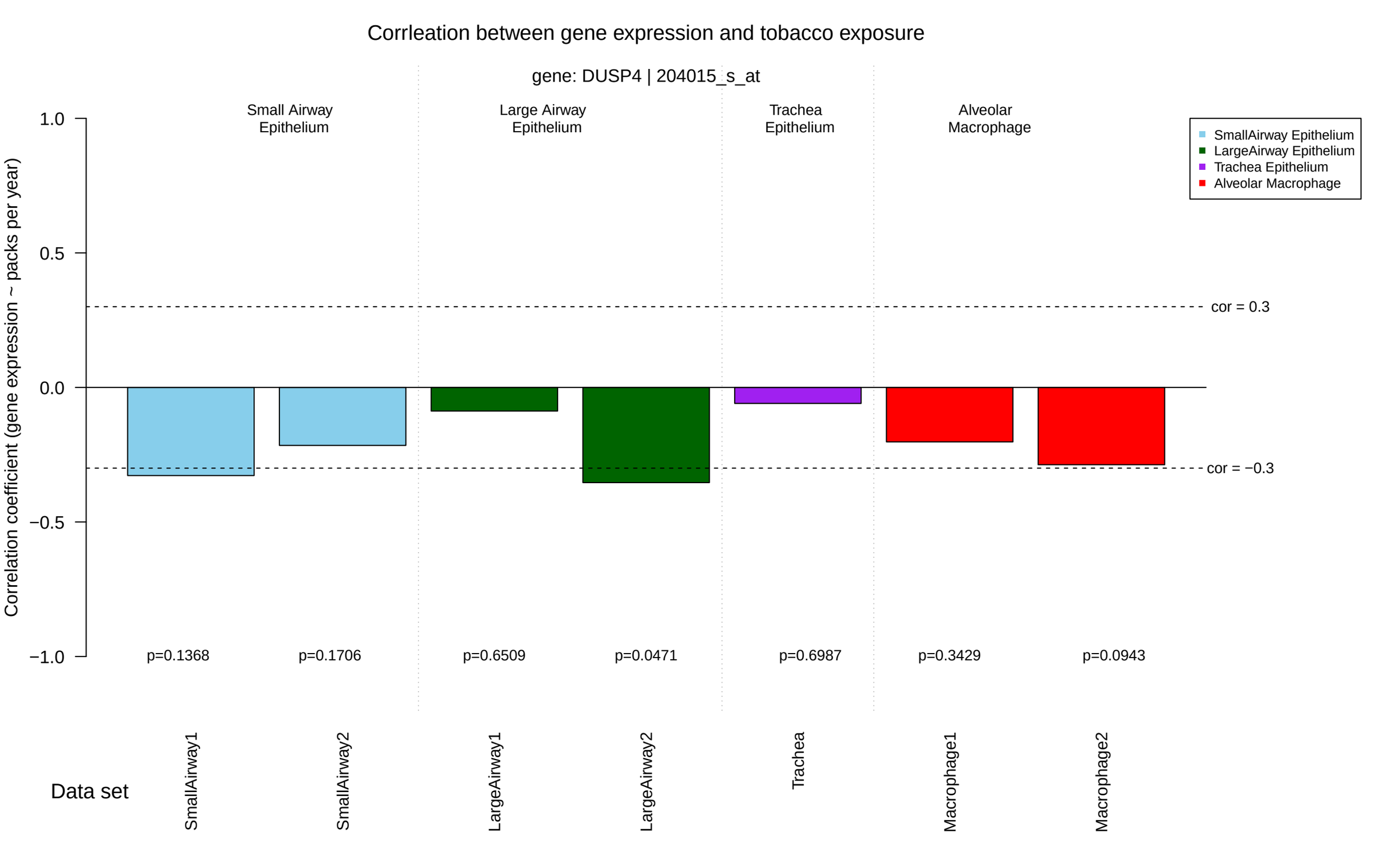

set: LargeAirway1 (p=0.339)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2859)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3286)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3168)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0075)
DUSP4 | 204015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DUSP4 | 204015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1368
p=0.1706
p=0.6509
p=0.0471
p=0.6987
p=0.3429
p=0.0943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2